

A 1.30^d Lower Bound for the Chromatic Number of Euclidean Space

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Abstract

We prove that $\chi(\mathbb{R}^d) \geq cC_*^d$ for an absolute constant $c > 0$ and all sufficiently large d , where $1.309251 < C_* < 1.309252$. In particular, $\chi(\mathbb{R}^d) \geq 1.30^d$ for all sufficiently large d . For infinitely many dimensions, the same construction gives $\chi(\mathbb{R}^d) \geq 1.316^d$. The construction uses all integer points with six or seven allowed coordinate values on a sphere centred at $(1/4, \dots, 1/4)$. The half-integer lattice containing their midpoints gives an elementary bound on their distances. A binary matrix has an identity principal submatrix on every colour class. Newton interpolation bounds its rank by a weighted sum of coefficients of a power of an explicit polynomial. One elementary binomial estimate supplies matching inverse-square-root factors in the vertex count and the rank bound, leaving a constant prefactor. Four concave lower bounds cover the interval forced by the choice of a power of two. The rounded bound also has a short proof using four integer comparisons with a common frequency denominator of 200. The algebraic arguments and all finite arithmetic data are included in the paper.

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1 The result and the proof

The chromatic number $\chi(\mathbb{R}^d)$ is the least number of colours needed to colour $\mathbb{R}^d$ so that points at distance one receive different colours.

Theorem 1. *There exist absolute constants $c > 0$ and d_0 such that*

$$\chi(\mathbb{R}^d) \geq cC_*^d \quad (d \geq d_0), \quad (1)$$

where C_* is defined by (21) and satisfies $1.309251 < C_* < 1.309252$. In particular, $\chi(\mathbb{R}^d) \geq 1.30^d$ for all sufficiently large d .

Raigorodskii's lower exponential base $1.239 \dots$ [6] is recorded in the 2025 survey [7, Section 1.5]. Theorem 1 concerns the same ordinary chromatic number and one forbidden Euclidean distance. Finite coordinate sets and linear algebra have important precursors in Frankl–Wilson [1]; entropy and diameter optimization appear in [2]. Related rank methods for multiple distances and forbidden graph copies are developed in [3, 5]. The one-coordinate algebra uses classical Newton interpolation for a Vandermonde matrix with geometric nodes; compare [4]. We prove the required divisibility in characteristic two.

The construction couples a quadratic layer about a quarter-integer centre with a coefficient rank bound in characteristic two. The centre gives the diameter bound, and the odd layer level ensures that $\|x - y\|^2/2$ is an integer for every pair of points on the layer. On the resulting nonempty set V , let G join pairs at one specified distance. After scaling, G is a unit-distance graph. A matrix $\mathcal{M}$ over $\mathbb{F}_2$ has an identity principal submatrix on every independent set I . Hence $|I| = \text{rank } \mathcal{M}[I, I] \leq \text{rank } \mathcal{M}$, and

$$\chi(\mathbb{R}^n) \geq \chi(G) \geq |V|/\text{rank } \mathcal{M}. \quad (2)$$

Keeping the entire layer gives matching inverse-square-root factors in the numerator and denominator, leaving a constant prefactor. Proposition 3 states the finite inequality. A choice between two coordinate sets makes its exponential base uniform in the dimension. Appendix A.1 gives a short route from that rate to 1.30^d , without locating the sharper constant.

2 A spherical layer and its detecting matrix

For $Q = 6$ or 7 , let $\mathcal{A}_Q$ consist of the first Q terms of

$$(a_0, \dots, a_6) = (0, 1, -1, 2, -2, 3, -3).$$

This orders the integers by increasing distance from $1/4$; both sets are consecutive integer intervals. Put

$$\tau(a) = 2a^2 - a, \quad T_j = \tau(a_j) = \frac{j(j+1)}{2}, \quad A_Q(t) = \sum_{j < Q} t^{T_j}.$$

For a power of two $P = 2^\ell$, with $\ell \geq 1$, define

$$V = \left\{ x \in \mathcal{A}_Q^n : \sum_{i=1}^n \tau(x_i) = 2P - 1 \right\}. \quad (3)$$

Equivalently, V is the intersection of $\mathcal{A}_Q^n$ with the sphere

$$\sum_{i=1}^n (x_i - 1/4)^2 = P - \frac{1}{2} + \frac{n}{16}. \quad (4)$$

For a series $F(u)$, write $[u^k]F(u)$ for its coefficient of u^k . We use all points on the layer, so

$$|V| = [t^{2P-1}]A_Q(t)^n. \quad (5)$$

Put $o = (1/4, \dots, 1/4)$. For $x, y \in V$, the midpoint $(x + y)/2$ lies in $(\frac{1}{2}\mathbb{Z})^n$, so every coordinate of $(x + y)/2 - o$ has absolute value at least $1/4$. The parallelogram identity and (4) give

$$\frac{1}{4}\|x - y\|^2 = P - \frac{1}{2} + \frac{n}{16} - \left\| \frac{x + y}{2} - o \right\|^2 \leq P - \frac{1}{2}. \quad (6)$$

Writing $R(x) = \sum_i x_i^2$ and $\rho(x, y) = \frac{1}{2}\|x - y\|^2$, we obtain

$$0 \leq \rho(x, y) \leq 2P - 1 \quad (x, y \in V).$$

Moreover, $R(x) \equiv \sum_i x_i \equiv \sum_i \tau(x_i) \equiv 1 \pmod{2}$. Thus $\rho(x, y) = (R(x) + R(y))/2 - \langle x, y \rangle$ is an integer, even though $R(x)$ can vary on V .

Let G join $x, y \in V$ when $\rho(x, y) = P$. Scaling by $(2P)^{-1/2}$ realizes G as a unit-distance graph in $\mathbb{R}^n$. Over $\mathbb{F}_2$, set

$$\mathcal{M}_{xy} = \binom{\rho(x, y) - 1}{P - 1} \pmod{2}. \quad (7)$$

For every integer h with $0 \leq h < 2P$, the binary coefficient rule gives

$$\binom{h - 1}{P - 1} \equiv 1 \pmod{2} \iff h = 0 \text{ or } h = P. \quad (8)$$

Indeed, for $h \geq 1$, expand $(1 + w)^{h-1}$ in characteristic two. Its coefficient of $w^{2^\ell - 1}$ is one exactly when all the lowest ℓ binary digits of $h - 1$ are one, which in this range means $h = P$. For $h = 0$, use $\binom{-1}{P-1} = (-1)^{P-1}$. Since $\rho(x, y) = 0$ only when $x = y$, the principal submatrix indexed by any independent set is the identity. Thus (2) applies.

Why powers of two? For any integer $P \geq 1$, let $f(h)$ equal one at $h = 0, P$ and zero elsewhere on $0 \leq h < 2P$. Its expansion $f(h) = \sum_{j < 2P} a_j \binom{h}{j}$ over $\mathbb{F}_2$ is unique, since the evaluation matrix is unit lower triangular. Binomial inversion gives

$$a_j = \sum_{h=0}^j \binom{j}{h} f(h) = 1 + \binom{j}{P} \pmod{2}.$$

If P is a power of two, the coefficients are one for $j < P$ and zero for $P \leq j < 2P$. Otherwise $a_{2P-1} = 1$: by the binary coefficient rule, $\binom{2P-1}{P}$ is odd exactly when the addition $P + (P - 1)$ has no carries, which occurs exactly for powers of two. Thus the least possible largest binomial index is $P - 1$ in the first case and $2P - 1$ otherwise. The same holds in the basis $\binom{h-1}{j}$, by an invertible triangular change of basis. This conclusion concerns the prescribed detector on the full integer window.

3 The rank estimate, one coordinate at a time

We work with formal power series in w over $\mathbb{F}_2$; only finitely many coefficients will be extracted. The order of a nonzero series is its least exponent with nonzero coefficient. A series with constant term one is invertible. Define

$$z = (1 + w)^{-1}, \quad e_0 = 0, \quad e_j = \sum_{h=1}^j 2^{\nu_2(h)}, \quad E_Q(s) = \sum_{j < Q} s^{e_j}, \quad (9)$$

where $\nu_2(h)$ is the exponent of two in h . In particular,

$$(e_0, \dots, e_6) = (0, 1, 3, 4, 8, 9, 11).$$

The T_j weight coordinate values in the vertex count; the e_j weight Newton basis elements in the rank bound. All matrix ranks are over $\mathbb{F}_2$. The coefficients of the counting polynomials A_Q, E_Q and their powers are ordinary nonnegative integers.

Lemma 2. *There exist matrices $U(w), W(w)$ with power-series entries such that*

$$B_Q(w) := (z^{ab})_{a,b \in \mathcal{A}_Q} = U(w) \operatorname{diag}(w^{e_0}, \dots, w^{e_{Q-1}}) W(w). \quad (10)$$

Proof. First take the coordinate set $\{0, \dots, Q-1\}$. Replace the monomials $1, X, \dots, X^{Q-1}$ by the monic Newton polynomials

$$f_0(X) = 1, \quad f_j(X) = \prod_{r=0}^{j-1} (X - z^r).$$

This is an invertible triangular change of basis over $\mathbb{F}_2[[w]]$. For $i < j$, $f_j(z^i) = 0$. For $i \geq j$, the order of $f_j(z^i)$ is

$$\sum_{r=0}^{j-1} 2^{\nu_2(i-r)} \geq e_j. \quad (11)$$

Here $z^h - 1$ has order $2^{\nu_2(h)}$: an odd exponent gives a nonzero linear term, and each squaring doubles the order. To prove the inequality in (11), use $2^{\nu_2(h)} = 1 + \sum_{k=1}^{\nu_2(h)} 2^{k-1}$ and the fact that every interval of j consecutive integers contains at least $\lfloor j/2^k \rfloor$ multiples of 2^k . These lower bounds are attained on $1, \dots, j$.

The j th column of the evaluation matrix $(f_j(z^i))$ is therefore divisible by w^{e_j} . Dividing these columns and undoing the basis change proves (10). Finally, if $\mathcal{A}_Q = \{L, \dots, L+Q-1\}$, then

$$z^{(L+i)(L+j)} = z^{L^2} z^{Li} z^{ij} z^{Lj}.$$

The additional factors are invertible scalar, row, and column factors. A reordering of the coordinate values is harmless. $\square$

We use the following coefficient rank bound. For a finite sum $F(w) = \sum_i w^{\lambda_i} u_i(w) v_i(w)^\top$, where u_i, v_i are vectors of series and λ_i, D are nonnegative integers,

$$\operatorname{rank}[w^D]F \leq \sum_i (D - \lambda_i + 1)_+. \quad (12)$$

Here $r_+ = \max(r, 0)$. Indeed, index i contributes

$$\sum_{r=0}^{D-\lambda_i} [w^r] u_i(w) [w^{D-\lambda_i-r}] v_i(w)^\top,$$

a sum of $(D - \lambda_i + 1)_+$ matrices of rank at most one. The sum is empty if $D < \lambda_i$. Row and column series factors can be absorbed into u_i, v_i before extracting coefficients.

Proposition 3 (Finite coefficient bound). *For $Q \in \{6, 7\}$, $n \geq 1$, and a power of two $P \geq 2$,*

$$\chi(\mathbb{R}^n) \geq \frac{[t^{2P-1}] A_Q(t)^n}{[s^{P-1}] (E_Q(s)^n / (1-s)^2)}. \quad (13)$$

Proof. If V is empty, the bound is immediate. Otherwise, the matrix $B_Q^{\otimes n}$ is indexed by $x, y \in \mathcal{A}_Q^n$ and has entry $\prod_i z^{x_i y_i}$. Tensoring (10) gives exponents $\lambda = e_{j_1} + \dots + e_{j_n}$, one for every tuple $(j_1, \dots, j_n) \in \{0, \dots, Q-1\}^n$.

For $x \in V$, put $d_x(w) = (1 + w)^{(R(x)-1)/2}$. The exponent is an integer because $R(x)$ is odd. We have the exact identity

$$(1 + w)^{\rho(x,y)-1} = d_x(w) \left(\prod_{i=1}^n z^{x_i y_i} \right) d_y(w). \quad (14)$$

The coefficient of w^{P-1} on the left is $\mathcal{M}_{xy}$. Restrict the tensor factorization to rows and columns in V , absorb d_x, d_y into its vector factors before extracting coefficients, and apply (12) with $D = P - 1$. This gives

$$\text{rank } \mathcal{M} \leq S_{n,Q}(P-1), \quad S_{n,Q}(D) = \sum_{k=0}^D (D-k+1) [s^k] E_Q(s)^n. \quad (15)$$

The right-hand sum is $[s^D](E_Q(s)^n/(1-s)^2)$. Equations (2) and (5) finish the proof. $\square$

4 A common coefficient estimate

We now compare the growth of the coefficients in (13). For a polynomial Z with nonnegative coefficients, the elementary bound $[u^m]Z(u)^n \leq Z(x)^n x^{-m}$, valid for $x > 0$, suggests the logarithmic rates

$$\begin{aligned} \Phi_Q(\mu) &= \inf_{0 < t < 1} \{\log A_Q(t) - \mu \log t\}, \\ \Psi_Q(\delta) &= \inf_{0 < s < 1} \{\log E_Q(s) - \delta \log s\}, \quad \Delta_Q(\delta) = \Phi_Q(2\delta) - \Psi_Q(\delta). \end{aligned} \quad (16)$$

For a list $b = (b_j)$ containing at least two distinct numbers, write

$$Z_b(x) = \sum_j x^{b_j}, \quad \mu_b(x) = \frac{\sum_j b_j x^{b_j}}{Z_b(x)}.$$

Here $\mu_b(x)$ is the mean exponent under the probabilities $x^{b_j}/Z_b(x)$. Its derivative with respect to $\log x$ is the corresponding positive variance. Thus the relevant minimizers are unique and satisfy

$$\mu_T(t_Q(\delta)) = 2\delta, \quad \mu_e(s_Q(\delta)) = \delta. \quad (17)$$

Throughout this section take $\delta \in K = [0.37, 0.75]$ and $Q \in \{6, 7\}$. Both minimizers stay in compact subintervals of $(0, 1)$, since 2δ and δ lie strictly between zero and the corresponding uniform means.

Both polynomials have the terms $1, u$. Once the other exponent choices are fixed, the remaining count is binomial. This gives the same $n^{-1/2}$ factor in both estimates.

Lemma 4 (Coefficient bounds). *Let $0 = b_0 < b_1 = 1 < b_2 < \dots < b_{q-1}$ be fixed integers, and let $J \subset (0, 1)$ be a compact interval. There are constants $c, C > 0, n_0$ such that, for $x \in J$ and $n \geq n_0$,*

$$cn^{-1/2} \leq \frac{x^m [u^m] Z_b(u)^n}{Z_b(x)^n} \leq Cn^{-1/2} \quad \text{if } |m - n\mu_b(x)| \leq 1. \quad (18)$$

Here m is an integer, and the upper bound holds for every integer $m \geq 0$.

Proof. Put $p_j = x^{b_j}/Z_b(x)$. Let $X_1, \dots, X_n$ be independent with $\Pr(X_i = b_j) = p_j$. The normalized coefficient in (18) is $\Pr(\sum_i X_i = m)$.

Set $\alpha = p_0 + p_1$ and $\theta = p_1/\alpha$. Record the positions and values of all X_i outside $\{0, 1\}$. If N positions remain and the recorded values sum to R , the remaining values, conditionally on this record, are independent Bernoulli variables with success probability θ . Their sum has the binomial distribution $\text{Bin}(N, \theta)$. The numbers $\alpha, \theta, 1 - \theta$ are uniformly bounded away from zero. Stirling's formula gives, uniformly for such θ ,

$$c_A k^{-1/2} \leq \Pr(\text{Bin}(k, \theta) = r) \leq C_0 k^{-1/2} \quad \text{if } |r - k\theta| \leq A\sqrt{k}, \quad (19)$$

for large k ; the upper bound holds for every integer r . To verify this, put $u = r/k$; in the indicated range, Stirling gives the logarithm of this probability as

$$-\frac{1}{2} \log k - k \left(u \log \frac{u}{\theta} + (1 - u) \log \frac{1 - u}{1 - \theta} \right) + O(1).$$

The bracket has value and derivative zero at $u = \theta$ and second derivative $1/(u(1 - u))$. It is therefore $O_A(k^{-1})$. The unrestricted upper bound follows by applying the same estimate at a mode, which is within distance one of $k\theta$.

Now $\mathbb{E}N = \alpha n$ and $\text{Var } N \leq n$. Also $W = R + \theta N$ is a sum of independent bounded variables: replace 0, 1 by θ in each X_i . Thus $\mathbb{E}W = n\mu_b(x)$ and $\text{Var } W \leq b_{q-1}^2 n$. Chebyshev's inequality gives $\Pr(N < \alpha n/2) = O(n^{-1})$. On its complement, the upper bound in (19) is $O(n^{-1/2})$; the exceptional probability is smaller. This proves the coefficient upper bound for every m .

If $|m - n\mu_b(x)| \leq 1$, put $\alpha_0 = \min_{x \in J} \alpha(x) > 0$ and $b_{\max} = b_{q-1}$. A union bound and Chebyshev's inequality give

$$\Pr\{N < \alpha n/2 \text{ or } |W - m| > B\sqrt{n}\} \leq \frac{4}{\alpha_0^2 n} + \frac{b_{\max}^2 + 1/n}{B^2}.$$

Choose $B^2 \geq 4(b_{\max}^2 + 1)$ and then $n \geq 16/\alpha_0^2$. With probability at least $1/2$, $N \geq \alpha n/2$ and $|(m - R) - N\theta| \leq B\sqrt{n} \leq B\sqrt{2/\alpha_0}\sqrt{N}$. For all sufficiently large n , (19) therefore makes the conditional probability at least $c_0 n^{-1/2}$. Averaging proves the lower bound. $\square$

Take $t = t_Q(\delta)$ and $s = s_Q(\delta)$, where $\delta = P/n \in K$. Lemma 4, applied to A_Q with $m = 2P - 1$, gives

$$|V| \geq c_1 n^{-1/2} t \exp(n\Phi_Q(2\delta)).$$

Apply its upper bound to every coefficient of E_Q^n in (15). Since $\sum_{r \geq 0} (r + 1)s^r = (1 - s)^{-2}$,

$$S_{n,Q}(P - 1) \leq C_1 n^{-1/2} \frac{s}{(1 - s)^2} \exp(n\Psi_Q(\delta)).$$

The factors $n^{-1/2}$ cancel. Compactness bounds $t(1 - s)^2/s$ away from zero, so Proposition 3 implies, uniformly for all sufficiently large n and powers of two P with $P/n \in K$,

$$\chi(\mathbb{R}^n) \geq c_Q \exp(n\Delta_Q(P/n)). \quad (20)$$

5 Four concave functions give the uniform base

The functions in (16) are determined by

$$\begin{aligned} A_6(t) &= 1 + t + t^3 + t^6 + t^{10} + t^{15}, & A_7(t) &= A_6(t) + t^{21}, \\ E_6(s) &= 1 + s + s^3 + s^4 + s^8 + s^9, & E_7(s) &= E_6(s) + s^{11}. \end{aligned}$$

Let $a_- = 371979/10^6$ and $a_+ = 371980/10^6$. The exact bounds in Appendix A show that there is a unique $a_* \in (a_-, a_+)$ with

$$\Delta_6(a_*) = \Delta_7(2a_*), \quad C_* := \exp \Delta_6(a_*). \quad (21)$$

They also give $1.309251 < C_* < 1.309252$ and $\log C_* < 0.27$.

The factor two in (21) comes from the choice of P . The least power of two not less than an satisfies $a \leq P/n < 2a$. We must therefore cover a whole interval $[a, 2a]$ by lower bounds for Δ_6 and Δ_7 .

For fixed $0 < s < 1$, put

$$G_Q(\delta; s) = \Phi_Q(2\delta) - \log E_Q(s) + \delta \log s. \quad (22)$$

The function $G_Q(\cdot; s)$ is concave, because Φ_Q is an infimum of affine functions. Also $G_Q(\delta; s) \leq \Delta_Q(\delta)$, with equality if $s = s_Q(\delta)$. A concave function is bounded below on an interval by the smaller endpoint value.

Write $s_6^* = s_6(a_*)$ and $s_7^* = s_7(2a_*)$. On the intervals with consecutive endpoints a_* , $21/50$, $11/20$, $17/25$, $2a_*$, use $G_6(\cdot; s_6^*)$, $G_6(\cdot; 2/5)$, $G_7(\cdot; 12/25)$ and $G_7(\cdot; s_7^*)$, respectively (Figure 1).

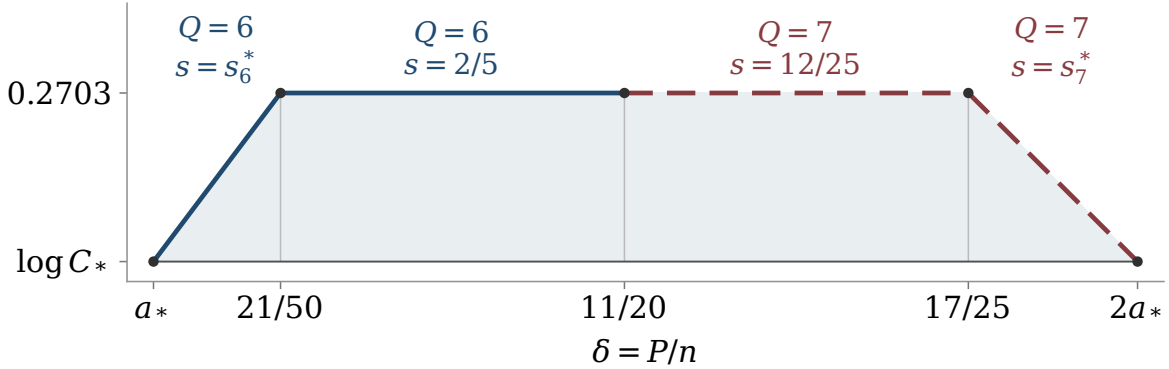


Figure 1: The four chords join proved endpoint lower bounds for the indicated functions $G_Q(\cdot; s)$. Concavity puts each function above its chord, and every chord stays at or above $\log C_*$.

The three interior cutpoints are auxiliary choices; they do not define C_* . At the two exterior endpoints the corresponding values equal $\log C_*$. Appendix A.2 proves that all six remaining endpoint values exceed $0.2703 > \log C_*$. Each concave function lies above the chord joining these endpoint lower bounds, so

$$\begin{aligned} \Delta_6(\delta) &\geq \log C_* & (a_* \leq \delta \leq 11/20), \\ \Delta_7(\delta) &\geq \log C_* & (11/20 \leq \delta \leq 2a_*). \end{aligned} \quad (23)$$

These two coordinate sets suffice for the bound; no claim of optimality among all coordinate sets is needed.

For each sufficiently large n , take the least power of two $P \geq a_* n$. Then $P/n \in [a_*, 2a_*) \subset K$. Use $Q = 6$ if $P/n \leq 11/20$ and $Q = 7$ otherwise. Combining (20) and (23), and taking the smaller of the two uniform constants, gives $\chi(\mathbb{R}^n) \geq cC_*^n$. Here the ambient dimension is n , as in (4); setting $n = d$ proves Theorem 1.

The need to cover a whole interval disappears if the dimension can be chosen along a sequence. A single rational parameter then gives a stronger base.

Corollary 5. *For infinitely many integers d , we have $\chi(\mathbb{R}^d) \geq 1.316^d$.*

Proof. Put $t_0 = 131/200$ and $\mu = t_0 A'_7(t_0)/A_7(t_0)$. Direct rational substitution gives $1 < \mu < 1.1$ and

$$A_7(t_0) > 2.03140 > 1.316 \cdot 1.54361 > 1.316 E_7(t_0^2).$$

For $P_\ell = 2^\ell$, choose n_ℓ to be a nearest integer to $(2P_\ell - 1)/\mu$. Then $n_\ell \rightarrow \infty$ and $|2P_\ell - 1 - n_\ell \mu| \leq \mu/2 < 1$. Choose $s = t_0^2$, so that $s^{P_\ell - 1}/t_0^{2P_\ell - 1} = t_0^{-1}$. Apply Lemma 4 at t_0 to the numerator of (13), and its unrestricted upper bound at s to each denominator coefficient. Summing the weights as in (15) gives, for some $c_0 > 0$ independent of ℓ and all sufficiently large ℓ ,

$$\chi(\mathbb{R}^{n_\ell}) \geq c_0 \frac{(1 - t_0^2)^2}{t_0} \left(\frac{A_7(t_0)}{E_7(t_0^2)} \right)^{n_\ell}.$$

The ratio is strictly greater than 1.316, so its exponential margin absorbs the constant prefactor for all sufficiently large ℓ . $\square$

A Finite rational bounds

All decimal numbers in this appendix denote exact rational numbers. We first give the four-row proof of 1.30^d , then certify the sharper constant. All arithmetic inputs are printed here.

A.1 Four integer comparisons give 1.30^d

Take $M = 200$, $C = 13/10$, and $s = u/M$. For nonnegative integers c_j with $\sum_j c_j = M$ and $\sum_j T_j c_j = 2K$, weighted arithmetic–geometric mean gives

$$A_Q(t)^M \geq \frac{M^M t^{2K}}{\prod_j c_j^{c_j}}, \quad 0^0 = 1.$$

Thus, with G_Q as in (22),

$$\begin{aligned} G_Q(K/M; u/M) - \log C &\geq M^{-1} \log \mathcal{R}, \\ \mathcal{R} &= \frac{M^M (u/M)^K}{C^M E_Q(u/M)^M \prod_j c_j^{c_j}}. \end{aligned} \tag{24}$$

Table 1 lists the inputs and the integer quotients $\lfloor 1000\mathcal{R} \rfloor$, obtained by integer multiplication and division after clearing denominators.

Table 1: Four integer witnesses with $M = 200$.

Q	K	$(c_0, \dots, c_{Q-1})$	u	$\lfloor 1000\mathcal{R} \rfloor$
6	75	(112, 63, 21, 4, 0, 0)	75	1850
6	100	(100, 64, 28, 7, 1, 0)	75	4288
7	100	(100, 64, 28, 7, 1, 0, 0)	96	1288
7	150	(86, 62, 33, 14, 4, 1, 0)	96	1391

Every last-column entry exceeds 1250, so $\mathcal{R} > 5/4$. Put $\gamma = M^{-1} \log(5/4) > 0$. Concavity gives $\Delta_Q(\delta) \geq G_Q(\delta; s) > \log C + \gamma$ on $[3/8, 1/2]$ with $(Q, s) = (6, 3/8)$, and on $[1/2, 3/4]$ with $(Q, s) = (7, 12/25)$. Taking the least power of two $P \geq 3n/8$ in (20) yields $\chi(\mathbb{R}^n) \geq c_0(Ce^\gamma)^n \geq 1.30^n$ for all sufficiently large n . Neither the crossing nor the logarithm enclosures below enter this argument.

A.2 The sharper constant

Logarithms and minimum values. For $x > 0$, let

$$\eta = \frac{x-1}{x+1}, \quad L(x) = 2 \sum_{j=0}^{15} \frac{\eta^{2j+1}}{2j+1}, \quad \varepsilon(x) = \frac{2|\eta|^{33}}{33(1-\eta^2)}. \quad (25)$$

Then $L(x) - \varepsilon(x) \leq \log x \leq L(x) + \varepsilon(x)$, by the geometric-series remainder. For accuracy, first write $x = 2^k y$ with $1 \leq y < 2$ and use $\log x = k \log 2 + \log y$, applying (25) to 2 and y . All resulting bounds are rational.

To enclose a minimum, suppose $\mu_b(x_-) < \mu < \mu_b(x_+)$ for $0 < x_- < x_+ < 1$. The function $f(y) = \log Z_b(e^y) - \mu y$ is convex, with derivative $f'(y) = \mu_b(e^y) - \mu$. Its minimum lies between $\log x_-$ and $\log x_+$. The tangent at the left endpoint therefore gives

$$f(\log x_-) - (\mu - \mu_b(x_-)) \log(x_+/x_-) \leq \min_y f(y) \leq f(\log x_-). \quad (26)$$

Indeed, the tangent has negative slope, so its value at $\log x_+$ is a lower bound throughout that interval.

The crossing. Put $N = 10^6$. Each row in the next table asserts

$$k_t/N < t_Q(\delta) < (k_t + 1)/N, \quad k_s/N < s_Q(\delta) < (k_s + 1)/N.$$

Table 2 gives open intervals for $10^9 \Delta_Q(\delta)$.

Table 2: Rational brackets for the crossing.

Q	δ	k_t	k_s	$10^9 \Delta_Q(\delta)$
6	0.371979	562503	342209	(269455401, 269455403)
7	0.743958	733755	517230	(269455447, 269455448)
6	0.371980	562504	342210	(269455480, 269455481)
7	0.743960	733755	517231	(269455367, 269455368)

For each t bracket, clear denominators in $\mu_T(k_t/N) < 2\delta < \mu_T((k_t + 1)/N)$; for s , use μ_e and δ . Apply (26) and (25) to obtain the last column.

The first two rows show $\Delta_6(a_-) < \Delta_7(2a_-)$, and the last two show the reverse inequality at a_+ . On this bracket, monotonicity of the means gives

$$t_6 < 0.563, \quad s_6 > 0.342, \quad t_7(2a) > 0.733, \quad s_7(2a) < 0.518.$$

Differentiation at the unique minima gives $\Delta'_Q(\delta) = \log(s_Q(\delta)/t_Q(\delta)^2)$. Since $0.342 > 0.563^2$ and $0.518 < 0.733^2$, $\Delta_6(a)$ is strictly increasing and $\Delta_7(2a)$ strictly decreasing. This proves existence and uniqueness in (21). The same log rule yields

$$\log(1.309251) < \frac{269455219}{10^9}, \quad \log(1.309252) > \frac{269455900}{10^9}.$$

Together with the first and third rows, these bounds prove the claimed enclosure for C_* and $\log C_* < 0.27$.

The six interior endpoint values. For each row below, $k/1000 < t_Q(\delta) < (k + 1)/1000$, as follows by clearing denominators in the two mean inequalities. Let $L_Q(\delta)$ be the lower bound for $\Phi_Q(2\delta)$ given by (26) with these endpoints. For every $s \in [s_-, s_+]$,

$$G_Q(\delta; s) \geq L_Q(\delta) - \log E_Q(s_+) + \delta \log s_-. \quad (27)$$

Table 3 records certified strict lower bounds for the right-hand side, obtained from (25). For example, its second row uses $t_- = 595/1000$, $t_+ = 596/1000$, $\mu = 21/25$ and $s = 2/5$ in (26) and (27).

Table 3: Six interior endpoint bounds. The last column bounds the expression in (27) strictly from below.

Q	δ	k	s_-	s_+	Lower bound
6	21/50	595	342209/ N	342211/ N	0.270365
6	21/50	595	2/5	2/5	0.270573
6	11/20	665	2/5	2/5	0.270652
7	11/20	664	12/25	12/25	0.270484
7	17/25	714	12/25	12/25	0.271274
7	17/25	714	517230/ N	517232/ N	0.270342

The crossing brackets and monotonicity imply that s_6^* and s_7^* lie in the intervals in the first and last rows. Since $0.2703 > 0.27 > \log C_*$, this proves all six strict inequalities used in Section 5.

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